

Architecture Overview

Architecture Part 1 - Email Triage

Email request -> Validate request -> Classify category -> Check critical keywords and seven-day contact frequency -> Critical: yes -> Route to human support -> Critical: no -> Retrieve relevant FAQ context -> Gemini classification and grounded drafting -> Deterministic fallback when Gemini is unavailable -> Return automated draft

Architecture Part 2 - Web Summarization

URL request -> Validate URL -> Fetch and parse webpage -> Remove scripts, styles, and boilerplate -> Normalize and limit source content -> Build guarded summarization prompt -> Gemini summary -> Validate maximum output length -> Return title, source URL, summary, and trimming status

Architecture Part 3 - Support Agent

User question -> Validate request -> Load session memory and FAQ document -> Gemini selects calculator, answer, or refuse -> Optional calculator tool -> Ground answer in FAQ content -> Deterministic local router when Gemini is unavailable or malformed -> Save latest conversation turn -> Return answer and tool metadata

Part	Trade-off	Pros and Cons	Potential Failure
1	A safety-first hybrid router applies deterministic escalation rules before AI classification and drafting.	Pros: Critical incidents and frequent contacts reliably go to a person. Drafts use retrieved FAQ context, and a fallback works without Gemini. Cons: Keyword rules can miss paraphrases or flag benign messages. Retrieval thresholds may reject relevant questions.	A critical issue without a known phrase may not be escalated. Missing FAQ context routes to a human. Bad Gemini output lowers classification quality but does not stop the fallback draft.
2	The system cleans and trims webpage text before sending it to Gemini instead of forwarding the complete page.	Pros: Reduces HTML noise, prompt size, and irrelevant content. Guardrails constrain input and output length. Cons: Cleaning can remove structure; trimming can omit details from long pages.	Invalid, private, or unreachable URLs are rejected. JavaScript-heavy pages may yield little content. Network or Gemini failures return an error rather than a summary.
3	A tool-aware support agent combines session memory, FAQ retrieval, model routing, and deterministic fallbacks.	Pros: Answers stay grounded, arithmetic uses a restricted calculator, and context supports follow-ups. It works without AI credentials. Cons: Memory is process-local and expires. Lexical retrieval may miss paraphrases. Model routing adds latency and cost.	Restarting or scaling can lose or split history. Only the latest 20 turns remain. Ambiguous arithmetic may parse incorrectly; unsupported factual questions are safely refused.